

Inventory management system

Workflow steps

step 1: data ingestion

- load cleaned dataset from:
 - delta table (/delta/inventory_management)
 - or csv (/filestore/inventory_csv)
 - data includes:
 - product name
 - warehouse name
 - current stock
 - reorder level
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step 2: data processing

- check for valid stock values
 - ensure stock is non-negative
 - handle missing values
 - validate data consistency
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step 3: filtering low stock products

- condition:

`current_stock < reorder_level`

- output:
 - list of products that need to be reordered
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step 4: report generation

- generate a csv report:
 - filename: daily_low_stock_report.csv
- columns:
 - product name
 - warehouse name
 - current stock

- reorder level

step 5: logging

- log pipeline execution:
 - start time
 - end time
 - number of products flagged
- example log:

pipeline started at 09:00 am
total products flagged: 4
pipeline completed successfully

Python script

```
import pandas as pd
from datetime import datetime

start_time = datetime.now()
print("pipeline started at", start_time)

df = pd.read_csv("inventory_csv")

df['current_stock'] = df['current_stock'].fillna(0)
df['current_stock'] = df['current_stock'].clip(lower=0)

low_stock = df[df['current_stock'] < df['reorder_level']]

low_stock.to_csv("daily_low_stock_report.csv", index=False)

end_time = datetime.now()

print("total products flagged:", len(low_stock))
print("pipeline completed at", end_time)
```

Simulated azure devops yaml pipeline

trigger:
- daily

pool:
vmImage: 'ubuntu-latest'

steps:

- script: echo "starting inventory pipeline"
displayName: "start"
- script: python inventory_report.py
displayName: "generate reorder report"
- script: echo "pipeline completed"
displayName: "end"